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AI-Powered Interview Preparation Coach

CAPSTONE PROJECT (CSEE-613 P)

A SYNOPSIS

Submitted to the

Department of Computer Science & Engineering
(Artificial Intelligence & Machine Learning)

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PROJECT SYNOPSIS

AI-Powered Interview Preparation Coach

A Multimodal AI System for Technical Interview Practice

Problem Statement:

Engineering students lack access to intelligent systems that provide comprehensive, multimodal feedback for technical interview preparation. Existing platforms focus solely on algorithmic problem-solving, neglecting critical evaluation of verbal articulation, non-verbal communication, and real-time response accuracy. There is a need for an AI-driven solution that simultaneously assesses technical proficiency, communication clarity, and behavioral cues to deliver holistic, actionable feedback for interview readiness.

1. Introduction

In today's competitive job market, engineering students face significant challenges in preparing for technical interviews. Traditional preparation methods often focus solely on theoretical knowledge or question practice, neglecting critical aspects such as communication skills and non-verbal cues that play a vital role in interview success. Our project addresses this gap by developing an AI-powered interview preparation system that provides comprehensive, multimodal feedback to help students excel in their technical interviews.

The AI-Powered Interview Preparation Coach is a web-based application that simulates real interview scenarios while analyzing three crucial dimensions simultaneously: technical accuracy of responses, verbal communication quality, and visual body language confidence. Unlike existing tools that offer generic feedback, our system leverages Retrieval Augmented Generation (RAG) to compare user responses against curated ideal answers, ensuring precise and contextually relevant evaluation.

This innovative platform empowers students to practice in a safe, judgment-free environment while receiving actionable insights that mirror real-world interview expectations. By integrating cutting-edge AI technologies including computer vision, natural language processing, and large language models, we aim to democratize access to high-quality interview coaching that was previously available only through expensive professional services.

2. Objectives

The primary objectives of this project are:

1. To develop a multimodal AI system capable of analyzing verbal responses, facial expressions, and body language in real-time during mock technical interviews.
2. To create an intuitive, user-friendly web interface that guides students through the complete interview preparation process with minimal technical barriers.
3. To deliver comprehensive, actionable feedback covering technical proficiency, communication effectiveness, and confidence indicators with personalized improvement recommendations.
4. To build a scalable and modular codebase that can be extended to support additional technical domains, interview types, and advanced analysis features.

3. Proposed Methodology

Our system employs a layered architecture that integrates multiple AI technologies to provide holistic interview assessment. The methodology is designed around five key processing layers that work in harmony to capture, analyze, and evaluate interview performance.

3.1 System Architecture

The system follows a multi-layered approach where each layer has specific responsibilities:

- **Presentation Layer:** Streamlit-based web interface that manages user interactions, session state, and role selection

- **Capture Layer:** Real-time acquisition of video (30 fps) and audio (5-second chunks) streams using OpenCV and microphone input.

- **Processing Layer:** Parallel processing pipelines for computer vision analysis (MediaPipe for facial landmarks and gaze tracking) and speech-to-text conversion (Whisper for accurate transcription).

- **Intelligence Layer:** RAG-based evaluation system using LangChain for orchestration, ChromaDB for vector storage, and large language models for semantic comparison and scoring.

- **Data Layer:** Curated database of technical interview questions with corresponding ideal answers, stored as embeddings for efficient semantic retrieval.

3.2 Data Flow and Processing Pipeline

The following diagram illustrates the complete data flow from user input to final feedback generation:

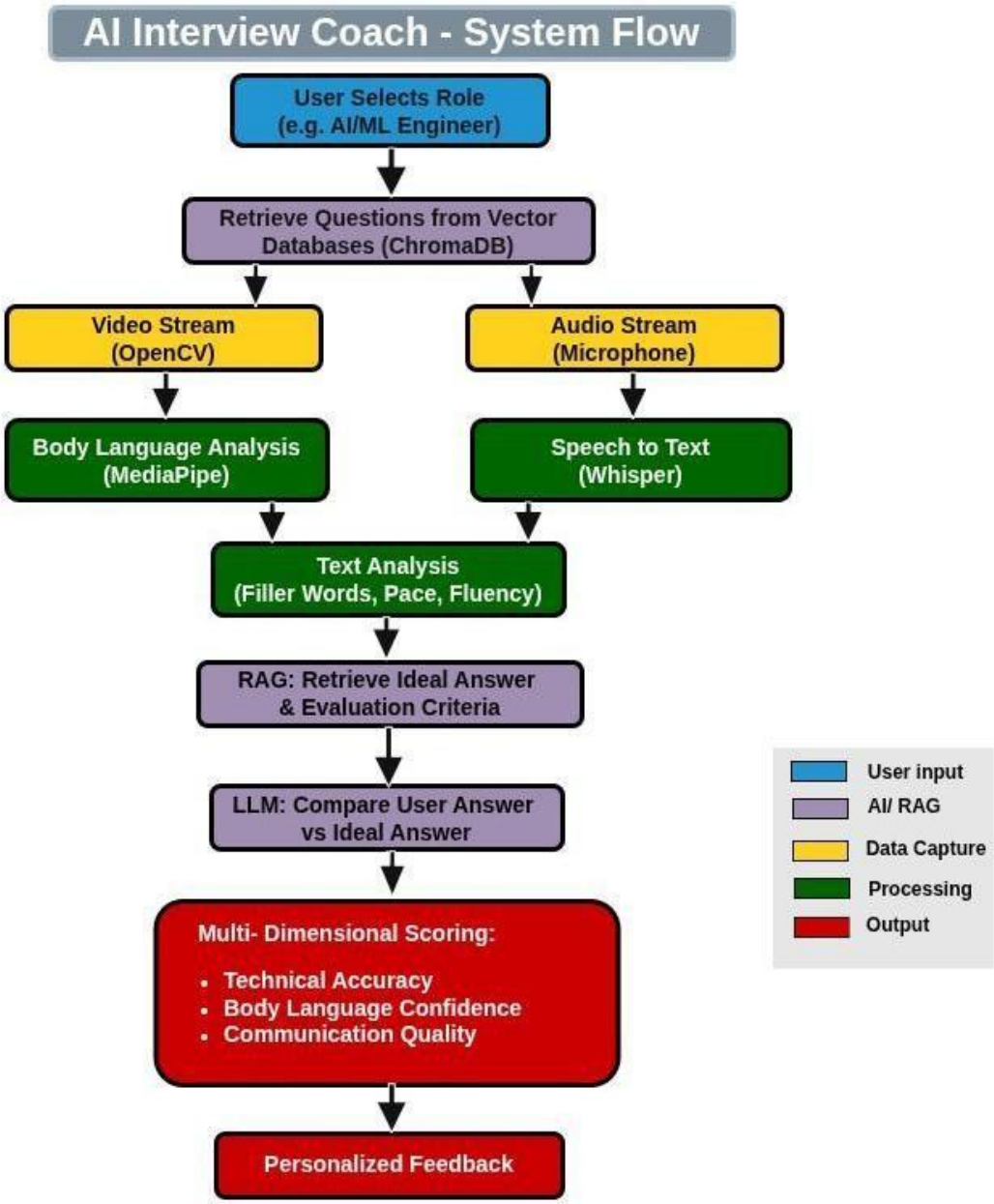

Figure 1: System Flow Diagram

The workflow begins when users select their target role (e.g., AI/ML Engineer, Data Scientist). The system then retrieves role-specific questions from ChromaDB vector database using semantic similarity search. As the user responds to each question, two parallel capture processes activate: video recording for body language analysis and audio recording for speech processing.

The captured data undergoes simultaneous processing where MediaPipe extracts facial landmarks to calculate gaze direction, eye contact percentage, and posture metrics, while Whisper converts speech to text with high accuracy. The transcribed text is then analyzed by NLP modules to detect filler words (um, uh, like), calculate speaking pace (words per minute), and assess fluency indicators.For technical evaluation, the RAG engine retrieves the ideal answer corresponding to the asked question from the vector database. A large language model then performs semantic comparison between the user's answer and the ideal answer, evaluating technical accuracy, completeness, clarity, and proper use of terminology.The system generates three distinct scores: Technical Proficiency (content accuracy), Body Language Confidence (eye contact and posture), and Communication Quality (pace, fluency, and clarity). These scores, along with specific improvement suggestions, are presented in an interactive dashboard.

4. Tools and Technologies

Our project leverages a modern, open-source technology stack carefully selected for scalability, performance, and ease of integration:

Category	Technology	Purpose
Web Framework	Streamlit	Interactive web interface with session management
Computer Vision	OpenCV, MediaPipe	Video capture and facial landmark detection
Speech Processing	OpenAI Whisper	High-accuracy speech-to-text conversion
AI Orchestration	LangChain	RAG workflow and LLM integration
Vector Database	ChromaDB	Embedding storage and semantic search
Language Model	Llama 3.1	Semantic comparison and evaluation
NLP Libraries	NLTK, spaCy	Text analysis and linguistic processing
Programming	Python 3.10+	Core development language

5. Expected Outcomes

Upon successful completion of this project, we expect to deliver the following outcomes:

- **Functional Prototype:** A fully operational web-based interview preparation system capable of conducting mock interviews for at least five technical roles (AI/ML Engineer, Data Scientist, Software Developer, DevOps Engineer, Data Engineer).
- **Comprehensive Dataset:** A curated database containing 200+ technical interview questions with corresponding ideal answers, structured and embedded for efficient retrieval.
- **Multimodal Analysis System:** Real-time processing capabilities that accurately analyze video (gaze tracking, posture), audio (speech-to-text), and text (NLP analysis) with less than 2 seconds latency per analysis cycle.
- **Accurate Evaluation Engine:** RAG-based assessment system achieving 85%+ alignment with expert evaluations on technical accuracy, demonstrated through validation testing with domain experts.
- **Interactive Dashboard:** User-friendly results interface displaying three-dimensional scores (Technical Proficiency, Communication Quality, Confidence) with specific, actionable feedback and personalized improvement recommendations.
- **Scalable Codebase:** Well-documented, modular code repository with clear separation of concerns, enabling easy extension to additional domains, interview types, and analysis features in future iterations.
- **Validated User Experience:** Positive usability metrics from beta testing, targeting a System Usability Scale (SUS) score above 70, indicating good user acceptance and interface intuitiveness.
- **Complete Documentation:** Comprehensive project documentation including technical architecture, API specifications, user guides, and detailed project report suitable for academic submission and future development reference.

This project will demonstrate the practical application of advanced AI technologies in solving real-world educational challenges, while providing students with a valuable tool that significantly improves their interview readiness and career prospects.

6. Work Flow and Timeline

The project execution is structured in four distinct phases over an 12-week period, with clearly defined milestones and deliverables:

Phase	Timeline	Key Deliverables
Phase 1: Foundation & Data Preparation	Week 1-Week 2	• GitHub repository setup with branching strategy • Dataset creation: 200+ questions with ideal answers • Basic Streamlit UI with role selection interface • Development environment configuration • ChromaDB schema design and initialization
Phase 2: Capture & Processing Modules	Week 2-Week 6	• Whisper integration for speech-to-text • OpenCV webcam capture implementation • MediaPipe facial landmark detection • Gaze tracking algorithm development • Basic NLP module for filler word detection
Phase 3: AI Intelligence Layer	Week 5-Week 9	• LangChain RAG pipeline implementation • LLM integration for answer evaluation • Scoring algorithms and feedback generation • Complete NLP suite (pace, fluency, sentiment) • Backend API endpoint development
Phase 4: Integration, Testing & Documentation	Week 8-Week 12	• End-to-end system integration • Results dashboard with visualizations • Comprehensive testing (unit, integration, E2E) • Performance optimization and bug fixes • Complete documentation and project report • Final presentation preparation

This project represents a significant step forward in applying artificial intelligence to enhance educational outcomes. By combining multiple AI technologies into a cohesive system, we aim to provide students with a comprehensive tool that not only tests their knowledge but also helps them develop the confidence and communication skills essential for interview success. Through systematic development and rigorous testing, we are committed to delivering a high-quality, production-ready application that demonstrates both technical excellence and practical value.